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a. $A - \lambda I = \begin{bmatrix} 1-\lambda & 2 & 1 \\ 2 & 4-\lambda & 0 \\ 0 & 0 & 1-\lambda \end{bmatrix}$

b. $\lambda I - A = \begin{bmatrix} \lambda-1 & -2 & -1 \\ -2 & \lambda-4 & 0 \\ 0 & 0 & \lambda-1 \end{bmatrix}$

c. $\det(A - \lambda I) = (1-\lambda)[(1-\lambda)(4-\lambda) - 2(2)]$

$= (1-\lambda)(\lambda^2 - 5\lambda + 4 - 4) = (1-\lambda)(\lambda^2 - 5\lambda) = \boxed{-\lambda^3 + 6\lambda^2 - 5\lambda}$

d. $\det(\lambda I - A) = (\lambda - 1)[(\lambda - 1)(\lambda - 4) - (-2)(-2)]$

$= (\lambda - 1)(\lambda^2 - 5\lambda + 4 - 4) = (\lambda - 1)(\lambda^2 - 5\lambda) = \boxed{\lambda^3 - 6\lambda^2 + 5\lambda}$

e. $\lambda^3 - 6\lambda^2 + 5\lambda = 0 \rightarrow \lambda(\lambda - 1)(\lambda - 5) = 0 \rightarrow \boxed{\lambda = 0, 1, 5}$

f. $\lambda = 0: \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $\lambda = 1: \begin{bmatrix} 0 & 2 & 1 \\ 2 & 3 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ $\lambda = 5: \begin{bmatrix} -4 & 2 & 1 \\ 2 & -1 & 0 \\ 0 & 0 & -4 \end{bmatrix}$

g. $\lambda = 0$: $R_2 \rightarrow R_2 - 2R_1$: $[2 - 2(1) \quad 4 - 2(2) \quad 0 - 2(1)] = [0 \quad 0 \quad -2]$

$\rightarrow R_2 \rightarrow R_2 + 2R_3$: $[0 \quad 0 \quad 0] \rightarrow R_1 \rightarrow R_1 - R_3$: $[1 \quad 2 \quad 0]$

$x_3 = 0, \; x_1 = -2x_2, \; x_2 = t \rightarrow x = t(-2, 1, 0)$

Basis: $\boxed{\{(-2, 1, 0)^T\}}$

$\lambda = 1$: $R_1 \leftrightarrow R_2 \rightarrow R_1 \rightarrow (1/2)R_1$: $[1 \quad 3/2 \quad 0] \rightarrow R_2 \rightarrow (1/2)R_2$: $[0 \quad 1 \quad 1/2]$

$\rightarrow R_1 \rightarrow R_1 - (3/2)R_2$: $[1 - 0 \quad 3/2 - (3/2)(1) \quad 0 - (3/2)(1/2)] = [1 \quad 0 \quad -3/4]$

$x_3 = t, \; x_1 = (3/4)t, \; x_2 = -(1/2)t \rightarrow x = t(3/4, -1/2, 1)$

Basis: $\boxed{\{(3, -2, 4)^T\}}$

$\lambda = 5$: $R_1 \rightarrow R_1 + 2R_2$: $[-4 + 2(2) \quad 2 + 2(-1) \quad 1 + 2(0)] = [0 \quad 0 \quad 1]$

$\rightarrow R_3 \rightarrow R_3 + 4R_1$: $[0 \quad 0 \quad 0] \rightarrow R_1 \leftrightarrow R_2 \rightarrow R_1 \rightarrow (1/2)R_1$: $[1 \quad -1/2 \quad 0]$

$x_3 = 0, \; x_1 = (1/2)x_2, \; x_2 = t \rightarrow x = t(1/2, 1, 0)$

Basis: $\boxed{\{(1, 2, 0)^T\}}$